AM115 In-class exercise: Sushi Restaurant

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Problem Statement

The owner of a sushi restaurant comes to you for advice. She wants to know how much fresh fish she should order each day. Fresh fish bought on day 1 can of course be used for sushi on that day and may also be refrigerated and used on day 2 but won't be up to snuff on day 3. The number of customers that come each day is a random number. The owner, from years of experience, knows roughly how the random number is distributed. Could you help her?

1 Problem 1

What kinds of questions would you ask her (then come up with the answers/guesses yourself)?

The questions I would ask her are...

1. How much are you selling the sushi for, and what are the units?
2. How much does the sushi cost, and what are the units?
3. Since you can refrigerate the sushi, how much can you hold, and is there a cost?
4. Do you have data on the number of customers? With the data, we might be able to approximate the demand.
5. How much sushi can you order? Are there constraints? The idea is to get a sense of $u(t)$, the control function.

2 Problem 2

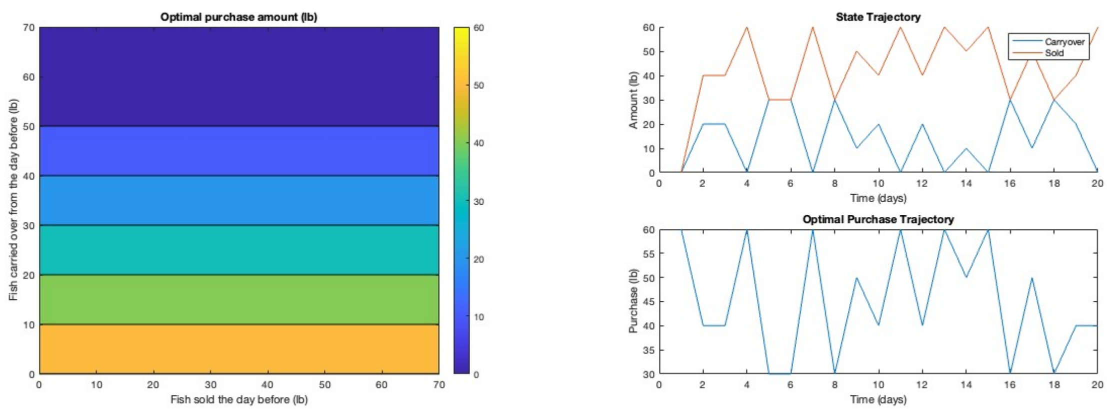
Formulate the problem: what should be the state variable(s), what is the control, what is the reward, how to evolve the state variable(s)? After you have formulated the problem, we will provide some code to solve it.

The state variables should be a vector, (amount leftover, amount sold). The control is the number of sushi to purchase in a given day, which is a function of carryover and the number of sushi sold the prior day. The reward is profit, which is the revenue minus the cost of fish and the holding cost. To evolve the state variable, we need to know the current state (carryover from yesterday, amount sold from yesterday), the demand, and the purchase amount.

3 Problem 3

Solve the optimal control problem using the sushi_ddp.m as the starting point.

Keeping all the parameters the same, running the MATLAB script gives us the following optimal control function, shown in Figure 1(A). Note that these contours are for $t = 1$. The contours do change at the end time step of $t = T$. The x-axis is the amount of fish sold in the prior day, and the y-axis is the carryover. We can apply this rule to the trajectories we observe in Figure 1(B). At $t = 8$, we sold 30 fish and had 30 carryover. Looking at the contours, we see the optimal policy would be to purchase 30 fish, which is also reflected in the optimal purchase trajectory. The last time step has an optimal purchase of 40, which is not the corresponding value in Figure 1(A). This is because the contours of the optimal purchase policy are different at $t = T$.



(a) (b)

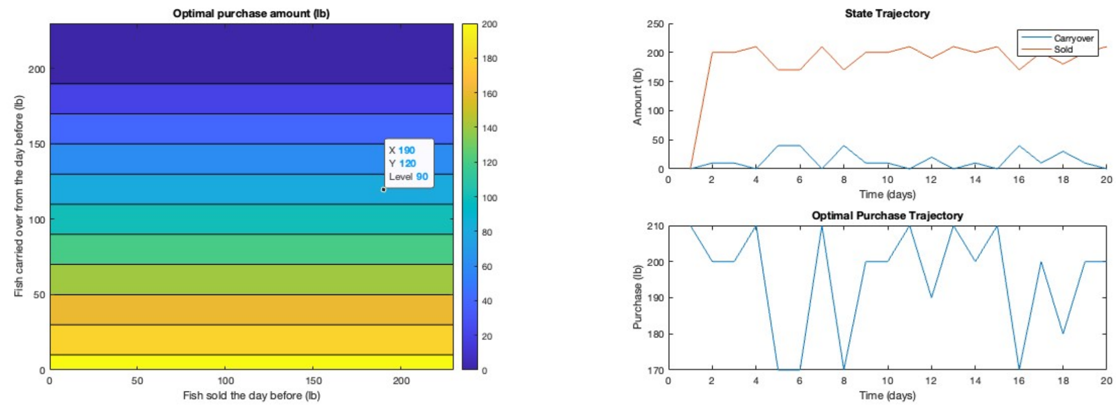
Figure 1: Optimal Control and Trajectories

4 Problem 4

What is the optimal purchase policy if this is big chain with many stores? What if this is a small business that's risk-averse and is content with making enough money to pay rent and other costs but will be under stress if that does not happen? In other words, how do we model risk? Modify sushi_ddp.m accordingly to find the optimal purchase policy in this case.

For a big chain with many stores, we make modifications to the parameters for `price`, `cost`, `holdcost`. These were all decreased due to economies of scale and a desire to set lower prices to undercut competition. The cost is lower because a large chain can buy wholesale at a bulk discount. The holding cost is lower because they can store in bulk, making it cheaper per unit. We also modify the probability distribution for demand to be concentrated at the middle value of the demand array. The idea is that we have more customers, and as you increase the sample size, by the law of large numbers, it converges to the population average. Finally, we expanded the set of possible demand from 5 to 7 and quadrupled the baseline demand to account for more customers.

The results are shown in the figures below. Since we have a larger demand, the trajectories of carryover and amount sold are not close together. There is also a larger set of actions to choose from due to the larger set of demand values. Note, it is difficult to see, but the optimal purchase is 210 when carryover is 0 (represented by the y-axis).



(a) (b)

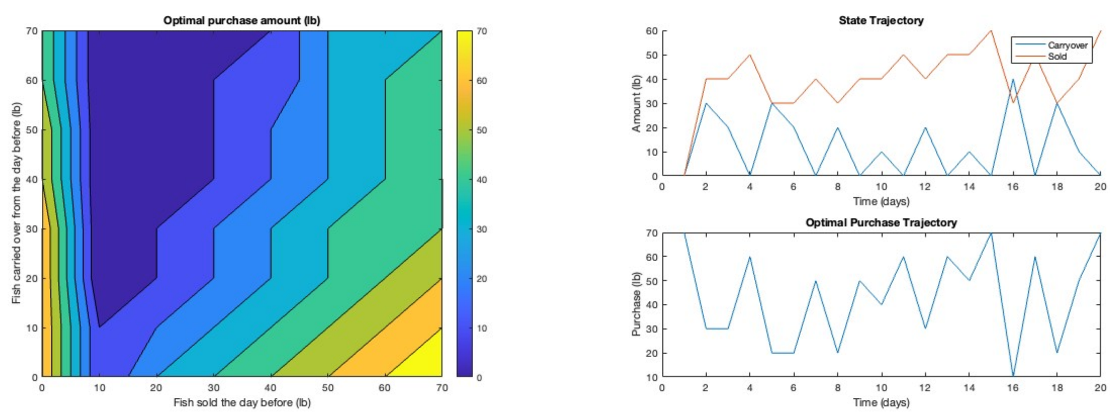
Figure 2: Optimal Control and Trajectories - Big Store

Now, we focus on the second scenario, where the store is small and risk-averse. Here, we revert the parameters to the default values. To model risk aversion, we use profit P as an input and use the function from the source linked:

$$U(P) = \frac{P^{1-\gamma} - 1}{1 - \gamma}. \tag{1}$$

The parameter γ controls risk aversion. If $\gamma > 0$, the store is risk-averse. $\gamma = 0$ is risk neural and $\gamma < 0$ is risk-seeking. We set $\gamma = 2.5$, which is quite strong risk-aversion.

After this modification, the results are shown in Figure 3. Looking at the contour plot in Figure 3(A), we see that the optimal purchase function has changed. It is no longer independent of fish sold the day before for a fixed carryover value. The purchase amount is now proportional to the fish sold the day before. If a lot of fish were sold, then we should purchase a lot, and vice versa. Like before, carryover is large, the optimal purchase decreases.



(a) (b)

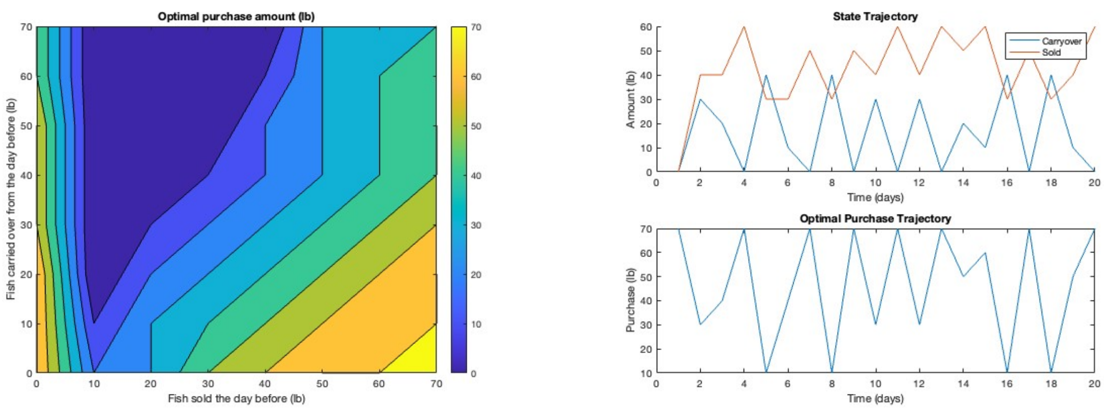
Figure 3: Optimal Control and Trajectories - Risk Averse

5 Problem 5

Modify the profit margin and other aspects of the problem to see how they affect the optimal purchase policy of the restaurant.

We keep the parameters the same as in the case of a small, risk-averse shop. Now, we increase the price to 60 and keep the cost at 10. This changes the profit margin from 10 to 50. It is admittedly exaggerated, but the point is to observe how the purchase contours and optimal trajectory change.

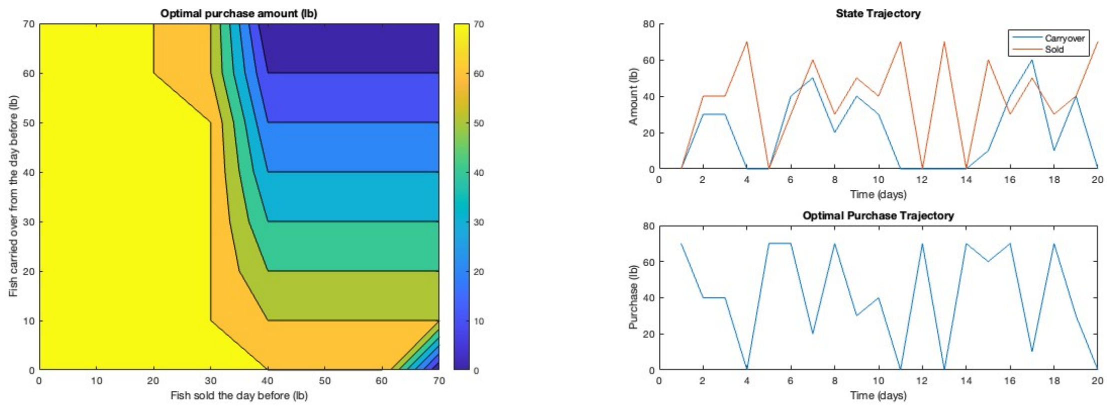
From Figure 4(A), we see that for a given fish sold and carryover, the optimal purchase is now higher than in Figure 3(A). This makes sense because we have higher profits and want to buy more fish to gain more profits. Many more points in the trajectory (Figure 4(B)) are at the maximum value of 70. This case is interesting because there are two strong opposing forces. The incentive for large profit and strong risk aversion. When there is no carryover, the shop buys as much as possible. When there is high carryover, the risk-aversion takes over, and the shop buys 0. This makes the variability of the purchase policy larger.



(a) (b)

Figure 4: Optimal Control and Trajectories - High Profit Margin

Finally, out of curiosity, we keep all parameters the same as Figure 3 but change the utility function to be risk-seeking by setting $\gamma = -1$. For a risk-seeking individual, if expected values are equal between two choices, they prefer to gamble because they derive extra utility from the possibility of large gains. As we suspect, the contours are essentially the opposite. When the shop has low sales, the risk-seeking store opts to purchase more, in hopes of getting more sales the next day. When carryover increases, the purchase amount decreases. Even though the store is risk-seeking, they still want to maximize profits.



(a) (b)

Figure 5: Optimal Control and Trajectories - Risk-Seeking

Sources

Utility Theory. (n.d.). https://www.worldscientific.com. Retrieved July 30, 2025, from `https://www.scribbr.com/citation/generator/folders/POoxzW6YC31o5pdYoNTh7/lists/QyssFpwHsJHklBeEHfMMJ/cite/webpage/`